

Quiz 1

Show all your work clearly. Write precisely. **Justify everything.** No credit for unsupported answers.

1. Consider the matrix $A \in \mathbb{R}^{3 \times 4}$ given by $A = \begin{pmatrix} 0 & 0 & 2 & 4 \\ 1 & 5 & 2 & 1 \\ 1 & 5 & 3 & 3 \end{pmatrix}$. Show that the Reduced Row Echelon Form (RREF) of A is $R = \begin{pmatrix} 1 & 5 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. Clearly specify the operation at each step.
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2. Solve the linear system $A\mathbf{x} = \mathbf{b}$, where A is the matrix from Problem 1 and $\mathbf{b} = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$. Describe the complete solution set S precisely. Then express S as a “translate of a subspace”, meaning you must specify a subspace W and a (constant) vector $\mathbf{p}$ such that $S = W + \mathbf{p}$, i.e., $S = \{\mathbf{w} + \mathbf{p} \mid \mathbf{w} \in W\}$.
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3. For general $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \mathbb{R}^3$, find necessary and sufficient condition(s) on b_1, b_2, b_3 for the linear system $A\mathbf{x} = \mathbf{b}$ to be consistent, where A is the matrix from Problem 1.
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4. Let U be the subspace of $\mathbb{R}^4$ spanned by the vectors $\mathbf{v}_1 = \begin{pmatrix} -12 \\ 1 \\ 14 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 15 \\ 0 \\ -23 \\ 1 \end{pmatrix}$. Find a matrix M , with as few rows as possible, such that U is precisely the null space of M (i.e., $U = \{\mathbf{x} \in \mathbb{R}^4 \mid M\mathbf{x} = \mathbf{0}\}$).

Hint: Look carefully at the given vectors and try to exploit some very helpful slots! (Note: You may realize that this problem is an instance of Problem 2b in Tutorial 2, which asked you for a method to find, for any given set of vectors in $\mathbb{R}^n$, such a “magic matrix”. That problem was stated as optional, and you do NOT need it here if you follow the hint.)

5. Let M be a $p \times q$ matrix with column vectors $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_q \in \mathbb{R}^p$ and row vectors $\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_p^T \in \mathbb{R}^q$. Prove the following from the definitions of span and linear independence.

- (a) Show that if $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_q$ are linearly independent, then $\text{span}(\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_p^T) = \mathbb{R}^q$.
- (b) Show that if $\text{span}(\mathbf{r}_1^T, \mathbf{r}_2^T, \dots, \mathbf{r}_p^T) = \mathbb{R}^q$ then $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_q$ are linearly independent.
- (c) So (the *rows* of any matrix are linearly independent) $\Leftrightarrow$ (its *columns* span the ambient space). How?

You may not directly appeal to any part of our table of equivalences, but you can of course recreate the logic that went into proving that table. You can freely use facts we developed before the table (in particular about solving a linear system of equations) but in each instance clearly state which fact you are using and how. Try to find the shortest path involving as few notions as possible. **The word *dimension* is not allowed**, so of course you may not use its invariance, even implicitly.

Solutions to Quiz 1¹

1. Solution: We start with the augmented-style representation or directly apply row operations to A :

$$A = \begin{pmatrix} 0 & 0 & 2 & 4 \\ 1 & 5 & 2 & 1 \\ 1 & 5 & 3 & 3 \end{pmatrix}$$

1. $R_1 \leftrightarrow R_2$:

$$\begin{pmatrix} 1 & 5 & 2 & 1 \\ 0 & 0 & 2 & 4 \\ 1 & 5 & 3 & 3 \end{pmatrix}$$

2. $R_2 \rightarrow \frac{1}{2}R_2$:

$$\begin{pmatrix} 1 & 5 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 5 & 3 & 3 \end{pmatrix}$$

3. $R_3 \rightarrow R_3 - R_1$:

$$\begin{pmatrix} 1 & 5 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}$$

4. $R_3 \rightarrow R_3 - R_2$:

$$\begin{pmatrix} 1 & 5 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

5. $R_1 \rightarrow R_1 - 2R_2$:

$$R = \begin{pmatrix} 1 & 5 & 0 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

2. Solution: Using the RREF matrix R from Problem 1 with the augmented column $\mathbf{b} = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$, we perform the identical row operations to get the reduced augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 5 & 0 & -3 & 2 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

This corresponds to the system of equations:

$$\begin{aligned} x_1 + 5x_2 - 3x_4 &= 2 \\ x_3 + 2x_4 &= 1 \end{aligned}$$

Choosing free variables $x_2 = s$ and $x_4 = t$ (where $s, t \in \mathbb{R}$), we express the basic variables as:

$$x_1 = 2 - 5s + 3t, \quad x_2 = s, \quad x_3 = 1 - 2t, \quad x_4 = t$$

In parametric vector form, the complete solution set S is:

$$S = \left\{ \begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -5 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 3 \\ 0 \\ -2 \\ 1 \end{pmatrix} \mid s, t \in \mathbb{R} \right\}$$

¹Let me know if you spot any errors. Writeup till the note in Answer 4 is by Gemini.

Expressed as a translate of a subspace $S = W + \mathbf{p}$:

$$\mathbf{p} = \begin{pmatrix} 2 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad W = \text{span} \left\{ \begin{pmatrix} -5 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ -2 \\ 1 \end{pmatrix} \right\}$$

3. Solution: For a general vector $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, applying the row operations from Problem 1 to the augmented matrix $[A \mid \mathbf{b}]$ yields the final row:

$$(0 \quad 0 \quad 0 \quad 0 \mid b_3 - b_2 - \frac{b_1}{2})$$

For the system to be consistent, the right-hand side of the zero row must be zero:

$$b_3 - b_2 - \frac{b_1}{2} = 0 \implies b_1 + 2b_2 - 2b_3 = 0$$

Thus, the necessary and sufficient condition for consistency is $b_1 + 2b_2 - 2b_3 = 0$.

4. Solution: Following the hint, we treat the given spanning vectors as the general solution of a homogeneous system in parametric form where $x_2 = s$ and $x_4 = t$ are free variables, and x_1, x_3 are pivot variables:

$$\mathbf{v}_1 = \begin{pmatrix} -12 \\ 1 \\ 14 \\ 0 \end{pmatrix} \implies \begin{cases} x_1 = -12 \\ x_2 = 1 \\ x_3 = 14 \\ x_4 = 0 \end{cases}, \quad \mathbf{v}_2 = \begin{pmatrix} 15 \\ 0 \\ -23 \\ 1 \end{pmatrix} \implies \begin{cases} x_1 = 15 \\ x_2 = 0 \\ x_3 = -23 \\ x_4 = 1 \end{cases}$$

Combining these with parameters s and t :

$$x_1 = -12s + 15t, \quad x_2 = s, \quad x_3 = 14s - 23t, \quad x_4 = t$$

Substituting $s = x_2$ and $t = x_4$ into the pivot equations gives:

$$x_1 = -12x_2 + 15x_4 \implies x_1 + 12x_2 - 15x_4 = 0$$

$$x_3 = 14x_2 - 23x_4 \implies -14x_2 + x_3 + 23x_4 = 0$$

Writing these equations in matrix form $M\mathbf{x} = \mathbf{0}$, we obtain the matrix with the minimum number of rows:

$$M = \begin{pmatrix} 1 & 12 & 0 & -15 \\ 0 & -14 & 1 & 23 \end{pmatrix}$$

(Note: Fewer rows would mean at most one equation, which would lead to at least three free variables. For a vector in U , knowing the value of x_2 and x_4 fixes the other two coordinates. That property is cannot be true for a solution set with more free variables.)

5. Let M be a $p \times q$ matrix with columns $\mathbf{c}_j$ and rows $\mathbf{r}_i^T$.

(a) Show that if $\mathbf{c}_1, \dots, \mathbf{c}_q$ are linearly independent, then $\text{span}(\mathbf{r}_1^T, \dots, \mathbf{r}_p^T) = \mathbb{R}^q$.

Solution: $M\mathbf{x} = x_1\mathbf{c}_1 + \dots + x_q\mathbf{c}_q$, so linear independence of the columns $\mathbf{c}_1, \dots, \mathbf{c}_q$ is equivalent to $M\mathbf{x} = \mathbf{0}$ having only the trivial solution, which, by our method to solve linear systems, is equivalent to every column of $\text{RREF}(M)$ containing a pivot.

RREF requires all entries to the left, below and above a pivot to be zero. Since each of the q columns contains a pivot, the RREF nature forces the top q rows of $\text{RREF}(M)$ to form the $q \times q$ identity matrix (with any remaining rows being all zeros). Thus the standard unit vectors $\mathbf{e}_1, \dots, \mathbf{e}_q$ in $\mathbb{R}^q$ appear as the first q rows in $\text{RREF}(M)$. As the standard unit vectors span $\mathbb{R}^q$ and since elementary row operations preserve row span, the original row vectors $\mathbf{r}_1^T, \dots, \mathbf{r}_p^T$ must also span $\mathbb{R}^q$.

- (b) Show that if $\text{span}(\mathbf{r}_1^T, \dots, \mathbf{r}_p^T) = \mathbb{R}^q$ then $\mathbf{c}_1, \dots, \mathbf{c}_q$ are linearly independent.

Solution: We prove this by contradiction. Suppose the columns $\mathbf{c}_1, \dots, \mathbf{c}_q$ are *not* linearly independent. This means the homogeneous system $M\mathbf{x} = \mathbf{0}$ has a non-trivial solution, which implies that when M is row-reduced, at least one column does not contain a pivot.

Let j be the index of a non-pivot column. Consider the standard basis vector $\mathbf{e}_j \in \mathbb{R}^q$, which has entry 1 in the j -th slot and 0 everywhere else (in particular in all slots corresponding to pivot columns). We claim that $\mathbf{e}_j$ is NOT in the span of the rows of $\text{RREF}(M)$ (proof below). But $\text{span of the rows of } \text{RREF}(M) = \text{span of the rows of } M$, which were given to span $\mathbb{R}^q$. This contradiction means every column of $\text{RREF}(M)$ must contain a pivot, forcing the columns of M to be linearly independent.

Proof of the claim: Consider any linear combination $\mathbf{w} = y_1 \mathbf{v}_1 + \dots + y_p \mathbf{v}_p$ of rows $\mathbf{v}_i$ in $\text{RREF}(M)$. Now for every nonzero $\mathbf{v}_i$, let k_i be the column in which row $\mathbf{v}_i$ has its pivot. By the nature of RREF, every other row of $\text{RREF}(M)$ has entry 0 in slot k_i . So the k_i -th entry of $\mathbf{w}$ is y_i . Now $\mathbf{e}_j$ has entry 0 in slot k_i , so if we want $\mathbf{e}_j = \mathbf{w}$, we must have $y_i = 0$. But then $\mathbf{w} = \mathbf{0}$. So $\mathbf{w} = \mathbf{e}_j$ is unattainable.

Another proof based on Arnav Nanal's solution: Take transpose of our slogan “ $A\mathbf{v} =$ linear combination of columns of A with weights in $\mathbf{v}$ ” to get a parallel slogan. Exploit both slogans by considering $\mathbf{y}^T M \mathbf{x}$. Using $\text{span}(\mathbf{r}_1^T, \dots, \mathbf{r}_p^T) = \mathbb{R}^q$, choose suitable vectors $\mathbf{y}^T$ that directly force $(M\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0})$ slotwise.

This slick proof did not use row reduction at all! Still, can you reconcile the two proofs?!

- (c) So (the rows of any matrix are linearly independent) $\Leftrightarrow$ (its columns span the ambient space). How?

Solution: Apply parts (a) and (b) directly to the transpose matrix M^T . The rows of M are the columns of M^T , and the columns of M are the rows of M^T , immediately yielding the equivalence.